

1. What is exception handling in Python?

Exception handling is a mechanism in Python used to handle errors that occur during the execution of a program.

An **exception** is an error that occurs while the program is running. For example, dividing a number by zero or trying to open a file that does not exist can cause an exception.

Example:

try:

```
a = 10
```

```
b = 0
```

```
print(a / b)
```

except:

```
print("Cannot divide by zero")
```

Output:

Cannot divide by zero

2. How do you use the try, except and finally blocks for handling exceptions in Python?

The try, except, and finally blocks are used together to handle exceptions in Python.

try block

The **try block** contains the code that may cause an exception.

except block

The **except block** handles the exception if an error occurs in the try block.

finally block

The **finally block** is executed whether an exception occurs or not. It is generally used for cleanup operations such as closing a file.

Example:

try:

 a = 10

 b = 0

 print(a / b)

except ZeroDivisionError:

 print("Cannot divide by zero")

finally:

 print("Program completed")